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| <b>Time</b>                | 1 hour and 30 minute examination including pre-released research focus material.                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Materials</b>           | Research focus material for each option will be pre-released as advance information <b>four</b> working weeks before the examination via the Edexcel website ( <a href="http://www.edexcel.org.uk">www.edexcel.org.uk</a> ).                                                                                                                                                                                                                                    |
| <b>Questions and marks</b> | <p>Students will be given a list of questions based on the <b>six</b> options. Students will select and answer <b>one</b> question that relates to the option they have studied. They will be required to write a long essay, in which they demonstrate and synthesise the results of their research.</p> <p>Each question is out of 70 marks.</p> <p>Students will <b>not</b> be able to take any pre-released or research materials into the examination.</p> |

## 4.3 Option 1: Tectonic Activity and Hazards

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| <b>Tectonic activity</b> | <p>Tectonic activity generates a wide range of natural hazards. The fundamental cause of these is plate tectonics, and thus the hazards have a distinct geography, linked to different tectonic settings. Tectonics is a key landscape-forming process which produces distinctive landforms in active regions, ranging from minor surface features such as faults or scarps to vast rift valleys and shield volcanoes.</p> |
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## Tectonic hazards

Tectonic hazards generate significant risk to human populations and their possessions, related to their vulnerability and the magnitude and frequency of hazardous events. Risk varies due to many factors including level of economic development, preparedness and education. Hazard impacts may be short term or long term. People respond to hazard risk in a variety of ways by, for instance, attempting to modify the hazardous event, through vulnerability or loss. Response depends on knowledge, technology and the availability of financial resources.

### Global synoptic context

#### Key linkages

##### Places

Where are the tectonically active regions of the world?

##### People

Who is affected by tectonic activity and its associated hazards?

##### Power

Who is responsible for managing the risk associated with tectonic hazards?

#### Suggested teaching and learning

Mapping plate boundaries and recent hazardous events.

Examining the socio-economic profiles of two countries at different levels of development, that have been affected by recent hazardous events.

Using a news website to draw up a mind map of organisations and groups involved in a well-known hazardous event, eg the 2005 Kashmir earthquake.

## 1 Tectonic hazards and causes

**Enquiry question:** *What are tectonic hazards and what causes them?*

| What students need to learn                                                                                                                                                                                                                                          | Suggested teaching and learning                                                                                                         |
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| ■ Tectonic hazards and disasters and what makes tectonic activity hazardous. There is a range of tectonic hazards associated with both volcanoes (lava, pyroclastics, ash, lahars, etc) and earthquakes (ground shaking, displacement, liquefaction, tsunamis, etc). | ■ Exploring the difference between a tectonic event, a hazard and a disaster and investigating why some tectonic activity is hazardous. |
| ■ Event profile of hazards, including frequency, magnitude, duration and areal extent.                                                                                                                                                                               | ■ Developing an awareness that hazards vary in their profile (frequency, magnitude, duration, areal extent, etc).                       |

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| <ul style="list-style-type: none"> <li>■ The causes of tectonic hazards, including the pattern of tectonic plates, their movements and possible causes of these movements.</li> </ul>                                                | <ul style="list-style-type: none"> <li>■ Investigating the global pattern of plates and their movements and researching current theories explaining these movements, largely relating to convection currents in the asthenosphere.</li> </ul> |
| <ul style="list-style-type: none"> <li>■ Tectonic activity associated with different types of plate margins (convergent, divergent and transform) and the impact of this on the spatial distribution of tectonic hazards.</li> </ul> | <ul style="list-style-type: none"> <li>■ Developing an awareness that specific tectonic activity occurs at different plate margins and that this affects the spatial pattern of tectonic hazards.</li> </ul>                                  |

## 2 Tectonic hazard physical impacts

**Enquiry question:** *What impact does tectonic activity have on landscapes and why does this impact vary?*

| What students need to learn                                                                                                                                                                                                                                    | Suggested teaching and learning                                                                                                                                                                               |
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| <ul style="list-style-type: none"> <li>■ The varying impact of extrusive igneous activity, including the formation of volcanic cones, fissures and lava plateaux.</li> </ul>                                                                                   | <ul style="list-style-type: none"> <li>■ Developing an awareness as to why extrusive igneous activity varies from place to place and investigating why this produces a range of landscape impacts.</li> </ul> |
| <ul style="list-style-type: none"> <li>■ The formation and morphology of different types of volcano and the characteristics of different types of eruption. How these variations link to the processes at different plate margins and at hot spots.</li> </ul> | <ul style="list-style-type: none"> <li>■ Investigating differences in the morphology and eruption characteristics of different types of volcano.</li> </ul>                                                   |
| <ul style="list-style-type: none"> <li>■ The varying impact of intrusive igneous activity, both major and minor.</li> </ul>                                                                                                                                    | <ul style="list-style-type: none"> <li>■ Examining the range of major and minor features of intrusive activity and how even intrusive activity can have impacts on the surface landscape features.</li> </ul> |
| <ul style="list-style-type: none"> <li>■ The effects that earthquakes can have on landscapes, such as fault lines, rift valleys and ground displacement, and the stresses and forces the ground is subjected to during such activity.</li> </ul>               | <ul style="list-style-type: none"> <li>■ Researching the varying impact of earthquakes, including analysis of the stresses giving rise to faults.</li> </ul>                                                  |

### 3 Tectonic hazard human impacts

**Enquiry question:** *What impacts do tectonic hazards have on people and how do these impacts vary?*

| What students need to learn                                                                                                                                                                  | Suggested teaching and learning                                                                                                                                         |
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| <ul style="list-style-type: none"><li>■ The reasons why people live in tectonically active areas and how this might relate to the level of economic development.</li></ul>                   | <ul style="list-style-type: none"><li>■ Investigating the varying reasons why people live in tectonically active areas (such as ignorance, choice, inertia).</li></ul>  |
| <ul style="list-style-type: none"><li>■ The range of hazards associated with different types of tectonic activity.</li></ul>                                                                 | <ul style="list-style-type: none"><li>■ Weighing up a range of impacts that tectonic hazards have on people, including physical, economic and social impacts.</li></ul> |
| <ul style="list-style-type: none"><li>■ The specific impacts of a range of tectonic hazards at a range of scales and at locations in countries at different stages of development.</li></ul> | <ul style="list-style-type: none"><li>■ Researching the range of impacts of hazards in contrasting locations.</li></ul>                                                 |
| <ul style="list-style-type: none"><li>■ Trends in frequency and impact over time.</li></ul>                                                                                                  | <ul style="list-style-type: none"><li>■ Developing an awareness of the varying patterns of impact over time, both long and short term (eg Park Model).</li></ul>        |